ECE501 Digital Image Processing

WEEK 7 PROGRESS REPORT

Multi-Patient Evaluation of Brain MRI Tumour Localization Using Classical Image Processing

Group 10 | BraTS2020 | FLAIR modality

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1. Week 7 Objective

In week 7, the test set is used to see if the current classical image-processing pipeline can detect the tumour-like regions in patients other than the development patient. Both the candidate segmentation (Week 5) and the largest connected component localization (Week 6) were compared based on the same binary segmentation ground-truth definition (whole tumour = seg > 0). Patient 001 is not a part of the primary evaluation averages and is treated as a development/reference case.

2. Summary of Previous Work

In week 5, candidate tumour regions were created by making use of FLAIR intensity processing, brain region mask, Otsu thresholding, morphological clean up and connected component analysis. This pipeline was used again in Week 6 and was designed to remove the largest area of the remaining connected components without reference to the ground truth. The Week 7 script applies min–max normalization, median filter of size 3 × 3, brain-mask extraction based on Otsu, thresholding using Otsu within the brain mask, opening and closing with radius 2, and selection of the largest area component with a minimum area of 50 pixels.

Note – the starting project brief includes CLAHE, but the provided Week 7 implementation doesn't include CLAHE. The report therefore refers to the implementation that can be seen in the submitted script and does not imply that CLAHE was used in this experiment.

3. Baseline Verification: Patient 001, Slice 67

According to the previous-results audit report, the dimensions of the FLAIR and segmentation volumes are 240×240×155, the extracted slice is 240×240, and the ground truth volume is labeled seg > 0, with ground truth defined as this value. It claims the biggest component was chosen based on area only and ground truth was applied only for evaluation after the prediction.

Method	**TP**	**FP**	**FN**	**Dice**	**IoU**	**Precision**	**Recall**
Week 5	4088	165	960	0.8790	0.7842	0.9612	0.8098
Week 6	4088	38	960	0.8912	0.8038	0.9908	0.8098

According to the supplied audit file, the recalculated counts and metrics match the previously reported Week 5 and Week 6 values (all listed checks pass). This confirms that the baseline as documented by that audit is correct. The report does not reprocess the NIfTI data from the source volumes, and it is the audit file which does the work for this verification.

4. Dataset and Patient Selection

One patient for development and reference and 4 patients for evaluation are reported. Patients evaluated were randomly selected, excluding Patient 001 and the 20 patient IDs that were previously mentioned in the project's slice_list.csv with NumPy random seed 42. The selected evaluation patients were BraTS20_Training_035, BraTS20_Training_167, BraTS20_Training_243 and BraTS20_Training_285.

Patient	**Group**	**Slice**	**Selection method**	**GT used for selection**
BraTS20_Training_001	dev	67	fixed_dev_slice	NO
BraTS20_Training_035	eval	61	max_bright_content_thresh0.3	NO
BraTS20_Training_167	eval	85	max_bright_content_thresh0.3	NO

BraTS20_Training_243	eval	66	max_bright_content_thresh0.3	NO
BraTS20_Training_285	eval	72	max_bright_content_thresh0.3	NO

This is a very important deviation from the original plan for which the script was submitted: in the submitted script, there are four new evaluation patients, each of which is processed on one of the four slices, instead of two slices per patient. The Patient 001 processing is restricted to the fixed reference slice 67; no other Patient 001 slice (non-reference) is included. For each evaluation slice, the script finds the slice in which the brightest FLAIR intensity occurs between 0 and 0.3. It does not employ the proposed ground-truth tumour-area eligibility rule (at least 200 pixels), and it does not select two slices that are spaced evenly between slices with tumours. These are adaptation errors on the methodological aspect and have to be reported.

5. Methodology

- Load the FLAIR and segmentation NIfTI volumes and pull the desired axial slice out.
- Take the axial slice repeatedly, and normalize to the min-max range.
- Apply a 3×3 median filter.
- Generate a brain-region mask with Otsu threshold procedure, Opening (disk radius equal to 3), hole filling, Largest connected component selection, closing and hole filling.
- Apply Otsu thresholding to pixels inside the brain mask to form candidate tumour pixels.
- Use opening and closing (disk radius = 2) operations, and filter connected components with size >= 50 pixels to remove the dirt from the candidate mask.
- Week 5 result: check the overall result of the union of all the retained parts.
- Week 6 result: choose the single largest retained piece in terms of area, no ground-truth is used for this selection.
- Once a prediction has been set, check each prediction against the whole-tumour reference mask (seg > 0).

6. Results

6.1 Per-patient results

Patient ID	Group	Slice	GT area (px)	W5 Dice	W5 IoU	W6 Dice	W6 IoU	W6 Prec.	W6 Recall
001	dev	67	5048	0.8790	0.7842	0.8912	0.8038	0.9908	0.8098
035	eval	61	2274	0.7666	0.6215	0.7979	0.6637	0.9409	0.6926
167	eval	85	797	0.3394	0.2044	0.4808	0.3165	0.3165	1.0000
243	eval	66	3410	0.9398	0.8864	0.9581	0.9196	0.9641	0.9522
285	eval	72	1156	0.3912	0.2432	0.7240	0.5674	0.6582	0.8045

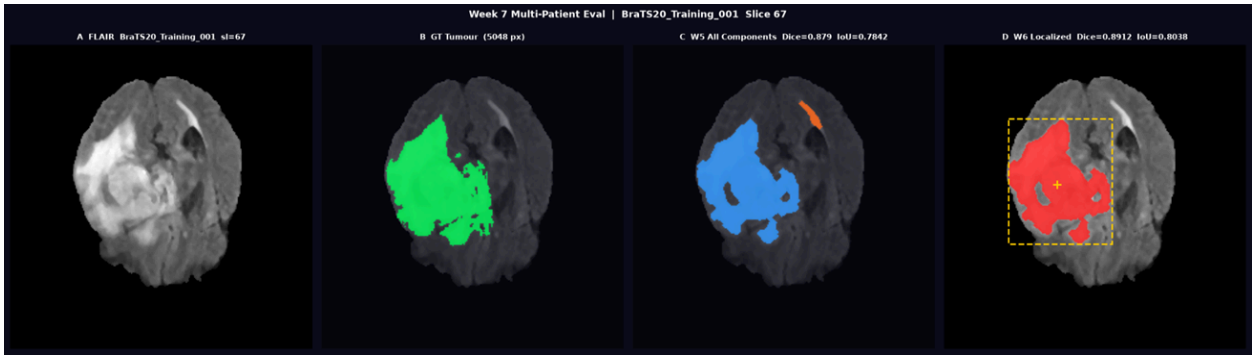
6.2 Primary evaluation-group aggregate

Method	Metric	Mean	Std. dev.	Min	Max
W5_all_comp	dice	0.6092	0.2522	0.3394	0.9398
W5_all_comp	iou	0.4889	0.2815	0.2044	0.8864
W5_all_comp	precision	0.5622	0.3322	0.2044	0.9277
W5_all_comp	recall	0.8623	0.1216	0.6926	1.0000
W6_localized	dice	0.7402	0.1720	0.4808	0.9581
W6_localized	iou	0.6168	0.2159	0.3165	0.9196
W6_localized	precision	0.7199	0.2622	0.3165	0.9641
W6_localized	recall	0.8623	0.1216	0.6926	1.0000

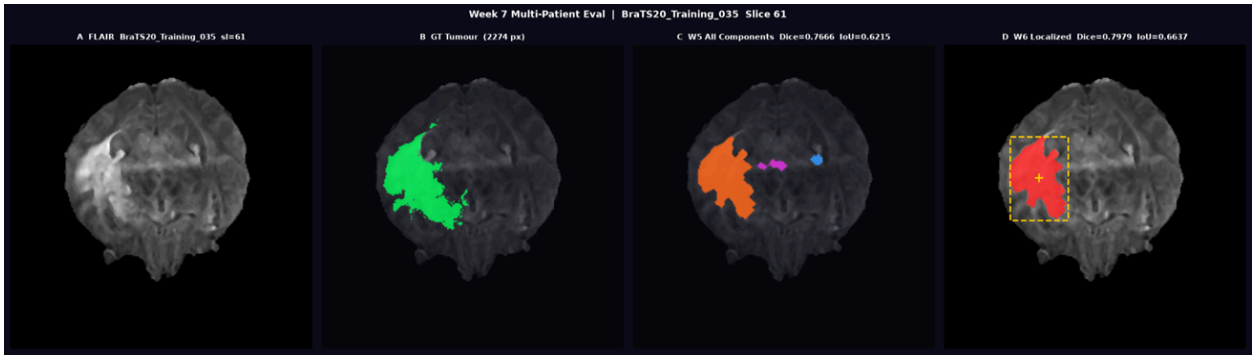
The primary evaluation has 6 patients, each of whom is assigned 1 selected slice (n = 4 slices). In the four cases, mean Dice ranged from 0.6092 at Week 5 to 0.7402 at Week 6, and mean IoU went from 0.4889 to 0.6168 in those four cases. Mean precision increased from 0.5622 to 0.7199. There was no difference in the mean recall, which was still 0.8623, with both methods. These are descriptive averages of four slices and not estimates based on a large or representative clinical group.

7. Visual Results

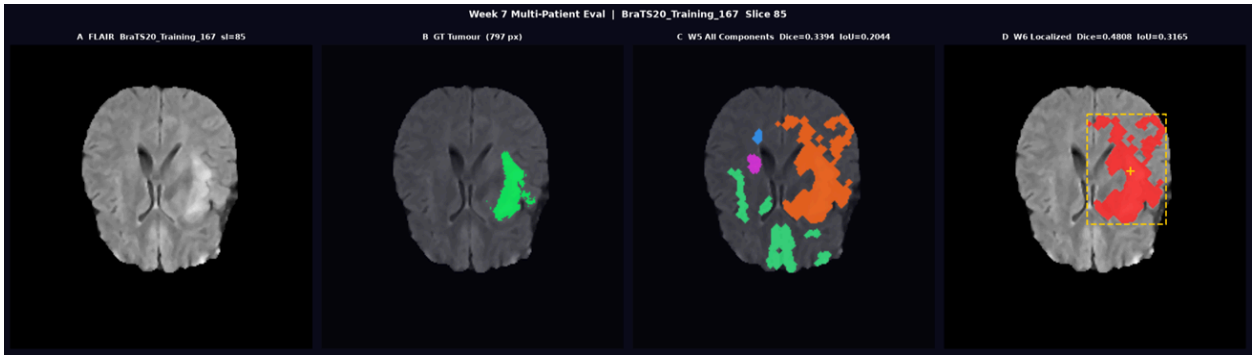
This set of comparison figures displays the FLAIR slice, the ground truth tumour mask, the FLAIR slice from the week 5 retained components and the FLAIR slice from the localized components computed at week 6. They visually represent the variation in performance: Patient 243 has high overlap, and Patient 167 has significantly less overlap. In patient 285 a significant improvement can be seen following the largest component selection, however, the mask is still not entirely representing the reference region.



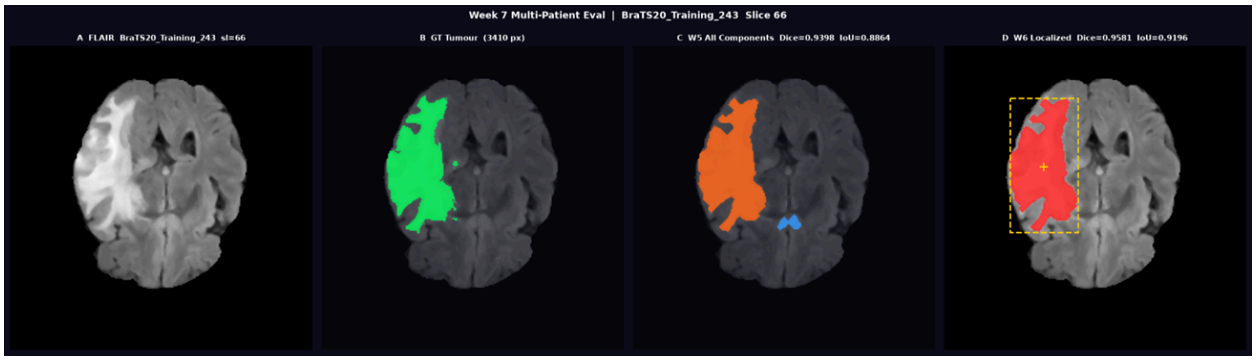
Development/reference case: Patient 001, slice 67.



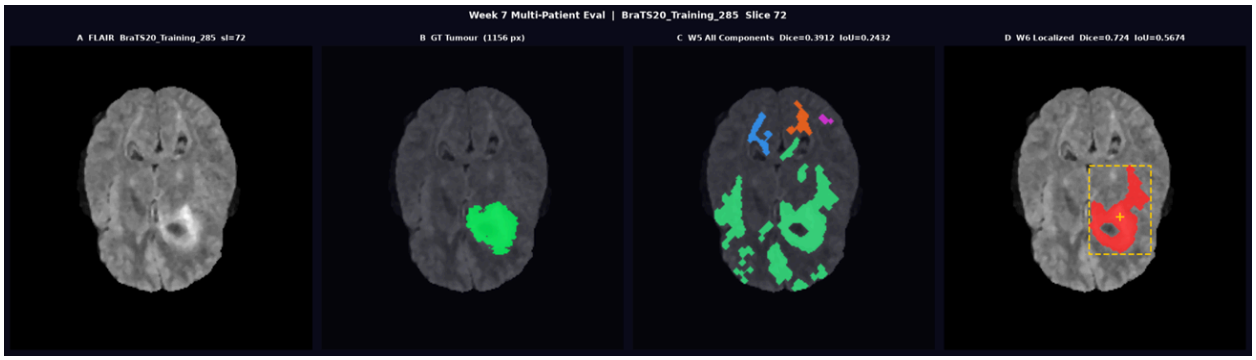
Evaluation case: Patient 035, slice 61.



Lower-performing evaluation case: Patient 167, slice 85.

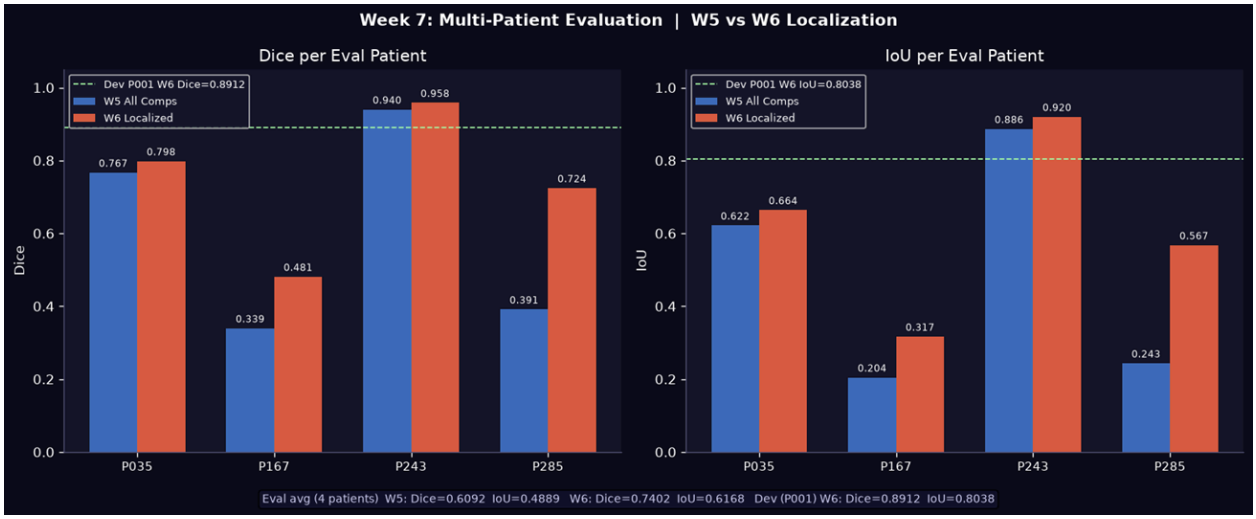


Higher-performing evaluation case: Patient 243, slice 66.



Evaluation case: Patient 285, slice 72.

7.1 Week 5 versus Week 6 comparison chart



8. Discussion and Observations

The biggest component rule increased the Dice and IoU on all four evaluation slices. The size of the improvement varied: Patient 243 was already strong at week 5 and made modest improvement; Patient 285 made a large improvement. Patient 167 was still the weakest patient following localization. The outcomes indicate that the elimination of smaller disconnected candidate components may help to decrease false-positive areas in these examples, while choosing just one component may be hazardous in the case of spatially distant tumour regions.

The overall results (aggregate) for recall is unchanged across the evaluation group (0.8623 for both methods), but the precision improves. This is in line with the highest component operation that removes the expected pixels outside of the reference mask, with no change in the mean percentage of recovered reference tumour pixels for these four cases. This is a statement about the metrics provided but not a guarantee that it will act the same on other patients.

The four evaluation slices contain ground-truth areas from 797 to 3410 pixels. Due to the small number of observations, it is not possible to draw a meaningful conclusion about the relationship between the localization performance and the area of the tumour. No CSVs were provided for centroid distance and tumour-coverage metrics, so they are not reported here.

9. Limitations and Methodological Notes

- A small exploratory sample (only 4 new evaluation patients and one slice per patient) was evaluated.
- The slice-selection rule that is implemented is not the same as the proposed two slices per patient, evenly spaced ground truth area qualified one.
- The selection rule is dependent on the number of bright normalized FLAIR pixels and can select slices for other reasons than that of tumour extent.
- The script submitted for Week 7 does not include any of the 20 IDs in the previous slice_list.csv, nor Patient 001; it is assumed that all other patients were not modified by any prior decisions of the parameters: it is recommended to check the full project history.
- Median filtering and Otsu/morphological operations used in code; CLAHE not used in provided Week 7 script even though it was mentioned in the brief.

- The largest component is the largest component of the bright retained component, which assumes that the largest retained bright component represents the main tumour region. This can break down if the non-tumour bright structures overpower or if the tumour areas are separated.
- The results are pixel-wise comparisons on selected 2D slices, not 3D tumour segmentation, clinical diagnosis or clinical validation.
- This run is not suitable to calculate per-patient mean metrics or a two-slice-per-patient analysis due to the fact that only one evaluation slice per patient was processed.

10. Conclusion

The results for the supplied Week 7 are a preliminary multi-patient test of the current classical image-processing pipeline. For the four selected evaluation patients, the largest-component localization at Week 6 achieved a higher mean Dice, IoU and precision than the all-component candidate mask at Week 5, with no change in the mean recall. The result was not consistent, with one evaluation case having a high overlap, but another case having a significant lower value. The results corroborate the usefulness of largest-component filtering for some cases, but do not provide strong generalization. Limitations of the method should be included in the conclusion of the final project.

11. Next Steps for Final Submission

- Consider whether to perform an evaluation of the two originally planned slices, or to keep the same run and label it as a one-slice-per-patient exploratory evaluation.
- Audit all four evaluation patients against the project's full development/tuning history, not the slice_list.csv listed above.
- If the output is not evaluated in the current output, preserve the existing output and save the current run as a new versioned run if it is evaluated in additional slices.
- Include component-level overlap, tumour coverage and centroid-distance if requested for localization analysis; only calculate these after predictions are fixed.
- Make an end-of-project flowchart and an end-of-project source code/run-instructions appendix.

- Do not use unverified numbers from the CSV in the final report or presentation; until verified, place names of group members and details of their contributions as placeholders.

Appendix A. Files used to prepare this report

- week7_multipatient_eval.py
- previous_results_verification.md
- week7_per_patient.csv
- week7_aggregate_summary.csv
- week7_patient_selection.csv
- week7_summary.md
- week7_summary_barplot.png
- Five supplied patient/slice comparison figures.